Session 5 — Advanced Probability & Statistics

Solution sheet

Complete worked solutions, with reminders of the key ideas. Numerical answers are boxed. Each solution keeps the concept in view — the calculation is there to make the idea concrete.

Exercise 1 *Prior Beta$(2, 2)$; 7 detections in 10 trials.*

Solution.

(a) Posterior $= \mathrm{Beta}(2+7,\ 2+3) = \boxed{\mathrm{Beta}(9,5)}$, with mean $\dfrac{9}{9+5} = \dfrac{9}{14} \approx \boxed{0.643}$.

(b) The prior mean was 0.5 and the data proportion 0.7; the posterior mean (0.64) sits *between* them — the data pulled the belief upward, but the prior still tempers it.

> **Reminder**
>
> The Beta prior is *conjugate* to binomial data: posterior $= \mathrm{Beta}(a + \text{hits},\ b + \text{misses})$, mean $\dfrac{a+h}{a+b+n}$. The prior acts like $a + b$ "pseudo-trials".

Exercise 2 *Then 3 more detections in 4 trials.*

Solution.

(a) From $\mathrm{Beta}(9, 5)$: add 3 hits and 1 miss to get $\boxed{\mathrm{Beta}(12,6)}$, mean $\dfrac{12}{18} = \boxed{0.667}$.

(b) All at once from $\mathrm{Beta}(2, 2)$ with 10 hits and 4 misses: $\mathrm{Beta}(2 + 10,\ 2 + 4) = \mathrm{Beta}(12, 6)$ — identical. This is *sequential updating*.

> **Reminder**
>
> Today's posterior is tomorrow's prior. Because Bayes multiplies the likelihoods, the order and grouping of the data do not change the final posterior.

Exercise 3 *Prior odds $1 : 3$, likelihood ratio 4 per firing.*

Solution.

(a) Posterior odds $= 4 \times \dfrac{1}{3} = \dfrac{4}{3}$, so $\mathbb{P} = \dfrac{4/3}{1+4/3} = \dfrac{4}{7} \approx \boxed{0.571}$.

(b) A second, independent firing multiplies the odds by 4 again: $\dfrac{16}{3}$, giving $\mathbb{P} = \dfrac{16}{19} \approx \boxed{0.842}$.

Reminder

Posterior odds = prior odds × likelihood ratio. Independent pieces of evidence simply *multiply* their likelihood ratios — the basis of evidence accumulation.

Exercise 4 *Joint distribution of S (present/absent) and R (yes/no).*

Solution.

(a) $\mathbb{P}(S{=}\text{pres}) = 0.35 + 0.05 = \boxed{0.40}$; $\mathbb{P}(R{=}\text{yes}) = 0.35 + 0.10 = \boxed{0.45}$.

(b) $\mathbb{P}(S{=}\text{pres} \mid R{=}\text{yes}) = \dfrac{0.35}{0.45} = \boxed{7/9 \approx 0.778}$.

(c) Not independent: $\mathbb{P}(S{=}\text{pres})\,\mathbb{P}(R{=}\text{yes}) = 0.40 \times 0.45 = 0.18 \neq 0.35$.

Reminder

Marginal = sum the joint over the other variable; conditional = joint / marginal; independence ⇔ joint = product of marginals.

Exercise 5 $\mathbf{\Sigma} = \left(\begin{smallmatrix} 4 & 2 \\ 2 & 4 \end{smallmatrix}\right)$.

Solution.

(a) $(4-\lambda)^2 - 4 = 0 \Rightarrow \lambda = \boxed{6 \text{ and } 2}$.

(b) Greatest-variance direction is the $\lambda = 6$ eigenvector, $(1, 1)$; it carries $\dfrac{6}{6+2} = \boxed{75\%}$ of the total variance.

(c) Not independent: the off-diagonal covariance is nonzero, and for a Gaussian "correlated" means "dependent".

Reminder

The covariance matrix links the two halves of the course: its eigenvectors are the principal axes of the data ellipse, its eigenvalues the variances along them.

Exercise 6 *Diagonal* $\mathbf{\Sigma} = \left(\begin{smallmatrix} 4 & 0 \\ 0 & 4 \end{smallmatrix}\right)$.

Solution.

(a) Yes, uncorrelated (the covariance is 0).

(b) Since the variables are jointly Gaussian, uncorrelated $\Rightarrow$ $\boxed{\text{independent}}$.

(c) Equal variances in both directions give *circular* contours (an isotropic Gaussian).

Reminder

"Uncorrelated $\Rightarrow$ independent" holds *only* for Gaussians. In general, zero covariance means no *linear* relationship, not independence.

Exercise 7 *Neuron responds on* 18 *of* 30 *trials, Bernoulli*(p).

Solution.

(a) $\ell(p) = k \ln p + (n-k)\ln(1-p)$ with $k = 18$, $n = 30$.

(b) $\hat{p} = \dfrac{k}{n} = \dfrac{18}{30} = \boxed{0.6}$.

(c) The logarithm is a strictly increasing function, so it has the *same* maximiser; it merely turns the product of trial probabilities into an easier sum.

Reminder

For Bernoulli/binomial data the MLE is just the observed proportion k/n — the formal machinery recovers the intuitive estimate.

Exercise 8

Solution.

(a) The Gaussian-mean MLE is the sample mean: $\dfrac{2+4+9}{3} = \boxed{5}$.

(b) Minimising squared errors is exactly maximising the likelihood when the noise is Gaussian — so least-squares regression *is* a maximum- likelihood method.

Exercise 9 8 *heads in* 10*: Model A* $(p = 0.5)$ *vs. Model B* $(p = 0.8)$.

Solution. The factor $\binom{10}{8}$ is common, so it cancels in the ratio:

$$\frac{0.8^8\, 0.2^2}{0.5^{10}} = \frac{0.006711}{0.000977} \approx \boxed{6.87}.$$

The data favour $\boxed{\text{Model B}}$, by a factor of about 6.9.

Reminder

A likelihood ratio compares how well two models predict the *same* data. Shared constants cancel, so only the model-specific factors matter.

Exercise 10 $\mathbb{P}(\textit{detect}) = \sigma(-1 + 2c)$.

Solution.

(a) At $c = 0.5$: $\eta = -1 + 2(0.5) = 0$, so $\mathbb{P} = \sigma(0) = \boxed{0.5}$.

(b) $\mathbb{P} = 0.5$ exactly when $\eta = 0$, i.e. $-1 + 2c = 0 \Rightarrow c = \boxed{0.5}$.

(c) $e^{\beta_1} = e^2 \approx \boxed{7.39}$: each unit of contrast multiplies the *odds* of detection by about 7.4.

Reminder

The sigmoid squashes any η into $(0, 1)$; the point $\eta = 0$ (probability 0.5) is the psychometric threshold, and e^{β} is the odds ratio.

Exercise 11 *Logistic coefficient $\beta = 0.7$.*

Solution.

(a) Odds ratio $e^{0.7} \approx \boxed{2.01}$.

(b) Since $\beta > 0$, the predictor *increases* the probability of the outcome.

(c) $\beta = \boxed{0}$ means no effect (odds ratio 1).

Exercise 12 $n = 200$; *M1* $(k = 3, \ln \hat{L} = -410)$, *M2* $(k = 6, \ln \hat{L} = -405)$.

Solution.

(a) $\text{AIC}_1 = 2(3) + 820 = \boxed{826}$; $\text{AIC}_2 = 2(6) + 810 = \boxed{822} \Rightarrow$ AIC prefers M2.

(b) $\text{BIC}_1 = 3(5.30) + 820 \approx \boxed{835.9}$; $\text{BIC}_2 = 6(5.30) + 810 \approx \boxed{841.8} \Rightarrow$ BIC prefers M1.

(c) BIC's per-parameter penalty $\ln n \approx 5.3$ is far larger than AIC's 2, so BIC punishes M2's three extra parameters much more heavily.

Reminder

$\text{AIC} = 2k - 2\ln \hat{L}$, $\text{BIC} = k \ln n - 2 \ln \hat{L}$. Both reward fit and penalise parameters; BIC penalises complexity more once $n > 7$. When they disagree, cross-validation breaks the tie.

Exercise 13 *Near-zero training error, poor held-out performance.*

Solution.

(a) This is $\boxed{\text{overfitting}}$: the model has high *variance* (it has fit the noise).

(b) Training error *always* falls as a model grows more flexible, so a low training error cannot, by itself, reveal overfitting.

(c) Cross-validation estimates *out-of-sample* (generalisation) error, which training error cannot see.

Reminder

The bias–variance trade-off: too simple $\to$ high bias (underfit); too complex $\to$ high variance (overfit). Judge models on held-out data.

Exercise 14 *Neuron with* $\mathbb{P}(\textit{spike}) = 0.25$.

Solution.

(a) $H = -0.25\log_2 0.25 - 0.75\log_2 0.75 = 0.25(2) + 0.75(0.415) \approx \boxed{0.811 \text{ bits}}$.

(b) Less than 1 bit: a biased (75/25) neuron is more predictable than a fair coin, so it carries less uncertainty.

(c) Entropy is maximal at $p = \boxed{0.5}$ (a fair coin, 1 bit).

Reminder

Entropy measures average uncertainty (in bits). It is largest for the uniform distribution and zero for a certain outcome.

Exercise 15

Solution.

(a) $I(S;R)$ measures how much knowing the response R reduces uncertainty about the stimulus S — the information they share.

(b) If R is independent of S, then $I(S;R) = \boxed{0}$.

(c) Since $D_{\mathrm{KL}} \geq 0$, cross-entropy $H(p,q) \geq H(p)$: describing the data with a *wrong* model q always costs extra bits, and can never beat the truth p.

Reminder

$I(X;Y) = H(X) - H(X \mid Y) \geq 0$, with equality only under independence. Minimising cross-entropy (as in training a classifier) drives q toward the true p.

Exercise 16 *Two attentional states; transition matrix as described; start* $(1,0)$.

Solution. Advance the state by multiplying with the transition matrix.

(a) After one trial: $(1)(0.8, 0.2) = \boxed{(0.8,\ 0.2)}$.

(b) After two: engaged $= 0.8(0.8) + 0.2(0.5) = 0.74$; distracted $= 0.8(0.2) + 0.2(0.5) = 0.26$, i.e. $\boxed{(0.74,\ 0.26)}$.

(c) *Markov property*: the next state depends only on the current state, not on the history — and advancing the distribution is just repeated multiplication by the transition matrix.

Reminder

A Markov chain propagates a probability distribution by matrix multiplication — another place where linear algebra and probability meet.